

Comments on: “Distinguishing orbiting and infalling dark matter particles with machine learning”. Draft by Ze’ev Vladimir

Edgar Salazar

April 1, 2025

From reading section 4.2, I think that there is a fine detail missing about what it means to compare SPARTA to ML and SPARTA to OASIS. I have written a general comment expressing my point of view. The reason I make emphasis in the comment below is because I have made similar mistakes before in some of my projects and talks, and oh boy that ruffled some feathers. You don't have to agree with me, but my intention is that you can keep this comments in mind when reviewing your draft. At the end of the day, you decide how to present your work. I am open whenever if something is not clear or just simply to discuss this further.

1 General comment

In principle there is no such thing as a *true* classification of orbiting and in-falling (O/I) particles at this point in time. I say this because we have no consensus on the operational definition. We do, however, have an agreed upon notion as to what it means for a particle (or any object for that matter) to be orbiting. I think it is any such object that has «accreted» into a larger structure and it is *now* orbiting the self generated potential. But what does it mean for an object to accret? Is a pericentric passage enough? For how long? What if it had an open orbit but the simulation stopped before it got out of the potential well? What if it got shot out by a merger or disruption? Of course, these and other issues arise when trying to implement (operationally)

this definition. You are free to decide which criteria to use and make an O/I split that is consistent with this physical notion.

That said, SPARTA is one choice of operational definition for O/I. Another choice was my [paper](#) with Rafael where we considered some accretion time and average radial velocity. Another choice is OASIS, which aims to be an optimisation over my previous work. Notice how I am using the word *choice* referring to the operational definition regardless of how useful or robust and algorithm is. When comparing between algorithms, one cannot use the word *accuracy* because none of the algorithms are accurate in the sense that there is no single true method. When you train an emulator or ML algorithm to mimic the results coming from one such operational definition like SPARTA and OASIS —because it is supervised learning you have to assume your training set to be the truth — it makes sense to talk about accuracy of the emulator or neural network.

I suggest you avoid talking about accuracy when comparing SPARTA and OASIS altogether. You can say things like “they do/don’t match”, “they do/don’t agree”. If you want a quantitative comparison between algorithms you can say something about how much they differ, e.g. “taking the ratio of the SPARTA mass to OASIS mass yeilds an average $\sim \delta\%$ difference or so”. Or “the number of parent haloes found by SPARTA and OASIS is X and Y , respectively”. When comparing density profiles, one can describe how the shape differs and try to explain why. But we cannot say which one is correct or wrong since there is no such thing. It is super important to understand the limitations of each algorithm so that we know when a comparison is fair or unfair.

2 Specific comments for section 4.2

1. The full OASIS algorithm includes percolation. This tells the algorithm how to classify substructure into O/I and also ensures that no particle is orbiting more than one structure. If I understand correctly, your results do not include this percolation step. If that is the case, I think it is important to explicitly state that you are comparing the labels you get by performing a cut in $\ln(v^2/v_{200m}^2) - r/R_{200m}$ space. Meaning that this is NOT OASIS. To get the true output and compare OASIS to SPARTA you need to run OASIS on the full dataset.

2. I will assume you are only using the calibration scheme from OASIS (not full OASIS) and refer to it as OASIS for convenience.
3. **Paragraph 2:** OASIS uses R_{200m} for calibration but R_{200b} for classification. This is crucial at low mass bins because R_{200b} is a better estimate (or at least I believe that to be the case) for the size of a lower particle count structure. However, for massive isolated haloes $R_{200m} \approx R_{200b}$.
4. **Paragraph 3:** It is not clear which optimal values you refer to from my paper, specially since you are calibrating the line yourself. I think you mean p and w . If so, I think it is better to rephrase that sentence to specify which parameters you are fixing and which you are fitting for and why. The calibration of m_p , b_p , m_n , b_n would not make sense otherwise.
5. **Paragraph 3:** If I remember correctly, you showed some plots of $\ln(v^2/v_{200m}^2) - r/R_{200m}$ for all boxes and we concluded that a cut of $1.75R_{200m}$ was good enough for the calibration. It is a technicality yes, but a difference in my algorithm because OASIS assumes $2R_{200m}$ as the cut for calibration and $2R_{200b}$ for classification. Since this differs from OASIS, you might want to explain why $1.75R_{200m}$ is chosen here.
6. **Paragraph 4:** The $1 h^{-1}$ Gpc box was used for calibration because it not only has more massive haloes with a large number of particles (resolution matters), but also because all the boxes have the same cosmology. Therefore you can use the same cut and not worry about cosmology dependence. I think this is not clear in the text.
7. **Paragraph 4:** The isolated halo criterion is with R_{200b} and not R_{200m} . Same with the mass threshold it is $0.2M_{200b}$ and not $0.2M_{200m}$.
8. **Paragraph 5:** The limit is $2R_{200b}$, not $3R_{200m}$.
9. **Figure 7:** The green line is too thin so it is hard to see. This is more of a suggestion but for colour-blind friendly plots, try to avoid greenish and reddish tones in the same plot. I think a black and thick line would do. You can also change the line style to dashed and repeat colours. The labels in all panels are too small. You might get corrections from referees if they are too small.

10. **Paragraph 5:** It is clear that SPARTA orbiting population has an uptick at low radii for $v_r < 0$. This is a notable difference between OASIS and SPARTA sure, but why is this? What makes it so that SPARTA considers those particles as orbiting and not in-falling?
11. **Paragraph 6:** This part discusses some drawbacks with OASIS. No issue with that since it is fair to talk about them. However, I do think that this related to my general comment. What does it mean that OASIS “struggles beyond R_{200m} in capturing the orbiting profiles’ dependence on mass in the transition region”? I suggest you revisit this discussion.
12. **Paragraph 7:** Again, it is not clear to me what do you mean by “both models are not trivially generalizable to other cosmologies”. I think you want to say that you cannot use the same cut lines you found for the Bolshoi boxes in the other two cosmologies. Is that the case? Regardless, I think the sentence sends a different message as it stands.

3 Specific comments in the rest of the paper

1. **Abstract or section 1:** It might be good to state that the ML algorithm is supervised learning so that the reader has a better understanding about the *truth* status of the O/I labels coming from SPARTA. Also, it may be good to include the name of the ML algorithm (XGBoost) in the abstract.
2. **Section 1, paragraph 3:** What do you mean by “Adding information such as the time when particles fell into a halo helps, but still does not fully separate the populations (Garcia et al. 2022)”.
3. **Section 2.1, paragraph 3:** Since the main product of the paper is the ML algorithm, SPARTA is by definition the “ground truth” for the training set.
4. **Figure 1:** In the caption you state that “A non-negligible fraction of orbiting particles exist outside of R_{200m} ”. It is not clear how large this fraction is. Perhaps give an numerical estimate? You could also add a legend for R_{200m} and $4R_{200m}$ in the figure so the reader does not have to go to the caption to find it.

5. **Section 2.2, paragraph 3:** Are you using physical velocities for section 4.2 as well? Please note that in the OASIS paper, the algorithm was developed for peculiar velocities. This could potentially impact the density profiles.
6. **Section 3.2:** Perhaps remind the reader that ML model was trained in Bolshoi cosmology, and that you are feeding data from WMAP7 and Planck cosmologies into the trained model. Also add references to those cosmologies or make a table with all 3 cosmologies so that it is easier to see how distinct these cosmological parameters are.
7. **Section 4.1:** I enjoyed this section.

Overall, cool paper. Congratulations!